

# Technical Report: Predicting Future Outcomes

## 1. Context

Turtle Games is a global manufacturer and retailer offering books, board games, video games, and toys. The company sells both its own products and those sourced from external suppliers, serving a diverse international customer base.

To improve sales performance, Turtle Games requires evidence-based insights into customer behaviour, loyalty engagement, and product feedback. This analysis explores loyalty point usage, customer segmentation, and review data, while also assessing the suitability of loyalty data for predictive modelling.

**Business Objective:** To improve overall sales performance by analysing customer trends, loyalty engagement, and reviewing data to guide marketing and product strategies.

## 2. Analytical Approach

The analysis was conducted using both Python and R to ensure a rigorous and comprehensive workflow, combining data wrangling, exploratory analysis, and predictive modelling to address Turtle Games' business objectives.

### 2.1 Data Preparation

The dataset, *turtle\_reviews.csv*, was first imported into Python. Libraries including pandas, NumPy, matplotlib, seaborn, and scikit-learn were used for cleaning and modelling, while nltk and wordcloud supported natural language processing tasks. Initial preparation involved examining data types, metadata, and missing values.

#### 1. Load and explore the data

```
[4]: # Imports
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import math
import re
import statsmodels.api as sm
import statsmodels.stats.api as sms
from statsmodels.formula.api import ols
from statsmodels.stats.outliers_influence import variance_inflation_factor
from collections import Counter
from nltk.probability import FreqDist
from nltk.corpus import stopwords
import sklearn
from sklearn import metrics
from sklearn import linear_model
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeRegressor, plot_tree
from sklearn.metrics import mean_squared_error, r2_score
```

Redundant columns such as *language* and *platform*, were removed to improve efficiency, and remaining variables were renamed to simpler forms (e.g., *remuneration* (*pay*), *spending\_score* (*spend*), *loyalty\_points* (*loyalty*)). The dataset was sense-checked to ensure no missing values, duplicate records were retained, considering the plausibility of the reviews having been written by customers and unnecessary columns were dropped for the convenience of analysis.

## 2. Drop columns

```
[10]: # Drop unnecessary columns.
reviews.drop(columns=['language', 'platform'], axis = 1, inplace = True)

# View column names.
reviews.columns

[10]: Index(['gender', 'age', 'remuneration (k£)', 'spending_score (1-100)',
          'loyalty_points', 'education', 'product', 'review', 'summary'],
          dtype='object')
```

## 3. Rename columns

```
[11]: # Rename the column headers.
reviews.rename(columns={'remuneration (k£)': 'pay',
                       'spending_score (1-100)': 'spend',
                       'loyalty_points': 'loyalty'}, inplace=True)

# View column names.
reviews.columns

[11]: Index(['gender', 'age', 'pay', 'spend', 'loyalty', 'education', 'product',
          'review', 'summary'],
          dtype='object')
```

The cleaned dataset was exported to CSV ('turtle\_reviews\_clean.csv') and re-imported for validation to ensure reproducibility and integrity.

## 4. Save the DataFrame as a CSV file

```
[12]: # Create a CSV file as output.
reviews.to_csv("turtle_reviews_clean.csv")

[13]: # Import new CSV file with Pandas.
reviews = pd.read_csv('turtle_reviews_clean.csv', index_col=0)

# View DataFrame.
reviews
```

|      | gender | age | pay   | spend | loyalty | education | product | review                                            | summary                                           |
|------|--------|-----|-------|-------|---------|-----------|---------|---------------------------------------------------|---------------------------------------------------|
| 0    | Male   | 18  | 12.30 | 39    | 210     | graduate  | 453     | When it comes to a DM's screen, the space on t... | The fact that 50% of this space is wasted on a... |
| 1    | Male   | 23  | 12.30 | 81    | 524     | graduate  | 466     | An Open Letter to GaleForce9*\n\nYour unpaint...  | Another worthless Dungeon Master's screen from... |
| 2    | Female | 22  | 13.12 | 6     | 40      | graduate  | 254     | Nice art, nice printing. Why two panels are f...  | pretty, but also pretty useless                   |
| 3    | Female | 25  | 13.12 | 77    | 562     | graduate  | 263     | Amazing buy! Bought it as a gift for our new d... | Five Stars                                        |
| 4    | Female | 33  | 13.94 | 40    | 366     | graduate  | 291     | As my review of GF9's previous screens these w... | Money trap                                        |
| ...  | ...    | ... | ...   | ...   | ...     | ...       | ...     | ...                                               | ...                                               |
| 1995 | Female | 37  | 84.46 | 69    | 4031    | PhD       | 977     | The perfect word game for mixed ages (with Mom... | The perfect word game for mixed ages (with Mom... |
| 1996 | Female | 43  | 92.66 | 8     | 539     | PhD       | 979     | Great game. Did not think I would like it whe...  | Super fun                                         |
| 1997 | Male   | 34  | 92.66 | 91    | 5614    | graduate  | 1012    | Great game for all.....\n\nKeeps the mind ni...   | Great Game                                        |
| 1998 | Male   | 34  | 98.40 | 16    | 1048    | PhD       | 1031    | fun game!                                         | Four Stars                                        |
| 1999 | Male   | 32  | 92.66 | 8     | 479     | PhD       | 453     | This game is fun. A lot like scrabble without ... | Love this game                                    |

2000 rows x 9 columns

```
[14]: # Sense-check the DataFrame
reviews.shape
```

```
[14]: (2000, 9)
```

## **2.2 Exploratory Data Analysis (EDA)**

EDA was performed in both Python and R. Summary statistics provided insights into distributions, skewness, and kurtosis, particularly for loyalty points, to test its suitability for predictive modelling. Visualisations such as histograms, scatterplots, and boxplots highlighted patterns in spend, pay, and loyalty point accumulation.

Correlation analysis helped to identify relationships between independent variables and loyalty points, while outliers were investigated to determine their potential business relevance.

## **2.3 Modelling and Advanced Analysis**

Several modelling approaches were applied. Linear regression and multiple linear regression were used to evaluate whether variables such as age, remuneration, and spending score could predict loyalty points.

Key parameters, error terms, and regression tables were extracted, and regression lines were plotted to interpret fit and reliability. To extend predictive capability, a Decision Tree Regressor was trained using a 70:30 train-test split, with pruning strategies applied to reduce overfitting.

Clustering techniques were employed to segment customers: remuneration and spending scores were analysed with k-means clustering, using the Elbow and Silhouette methods to determine the optimal  $k$ . This enabled clear identification of customer groups and their relative sizes.

## **2.4 Sentiment Analysis**

Customer reviews and summaries were cleaned through case conversion, punctuation removal, and tokenisation. Word clouds and frequency distributions highlighted common terms, while sentiment polarity was calculated to distinguish positive and negative reviews. These insights offered direct evidence of customer satisfaction and areas for improvement.

## **2.5 Rationale**

Each method was chosen to align directly with Turtle Games' objectives:

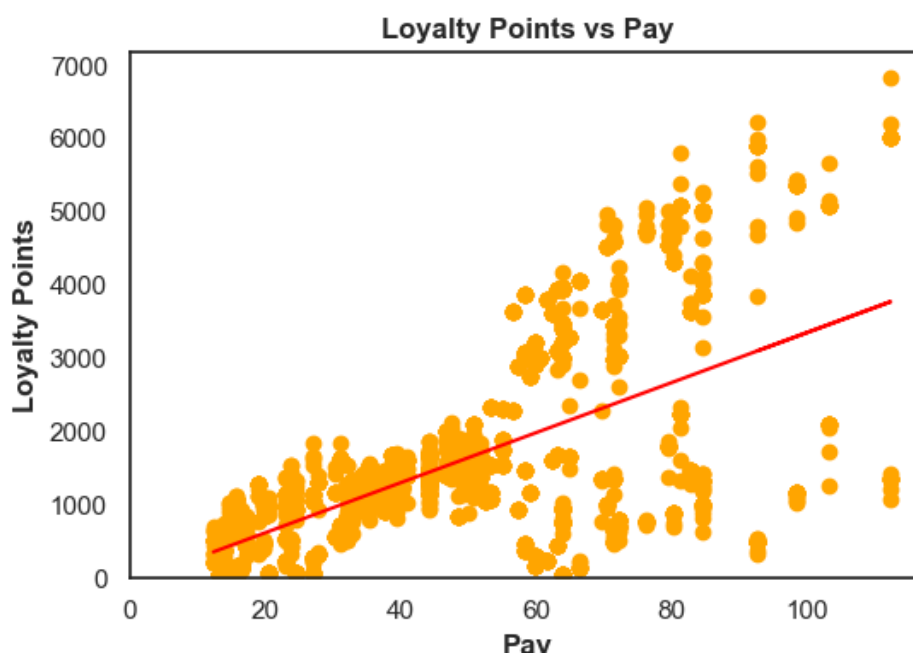
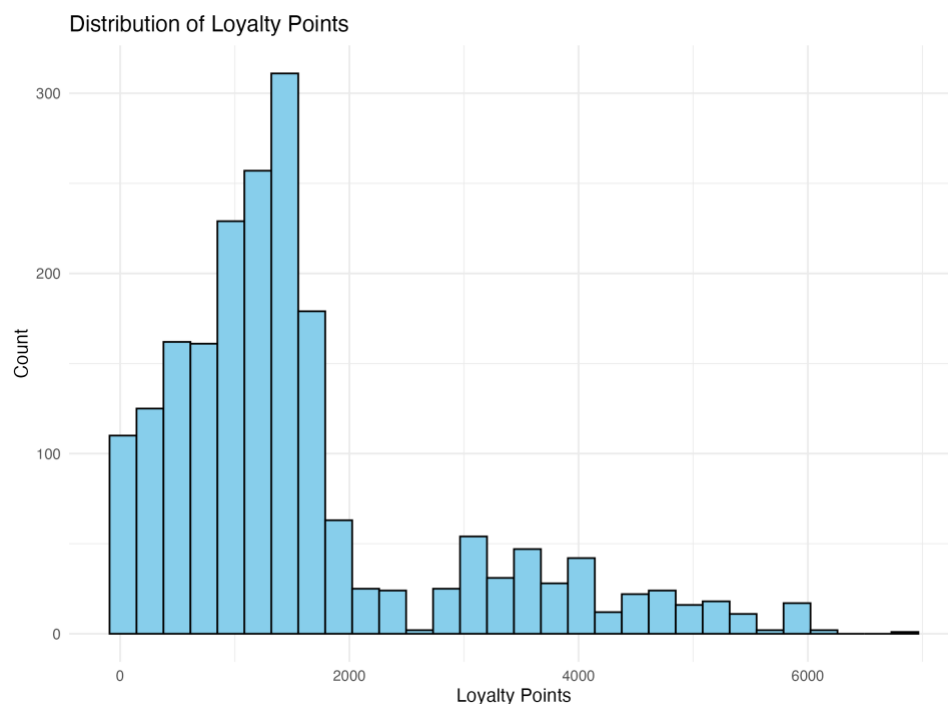
- Regression and decision trees tested the predictive power of loyalty data.
- Clustering provided a basis for targeted marketing.
- Sentiment analysis offered direct evidence of customer experiences.

This structured and transparent workflow ensured that findings were actionable, reproducible, and relevant to business needs.

### 3. Visualisation and Insights

#### 3.1 Loyalty Points Engagement

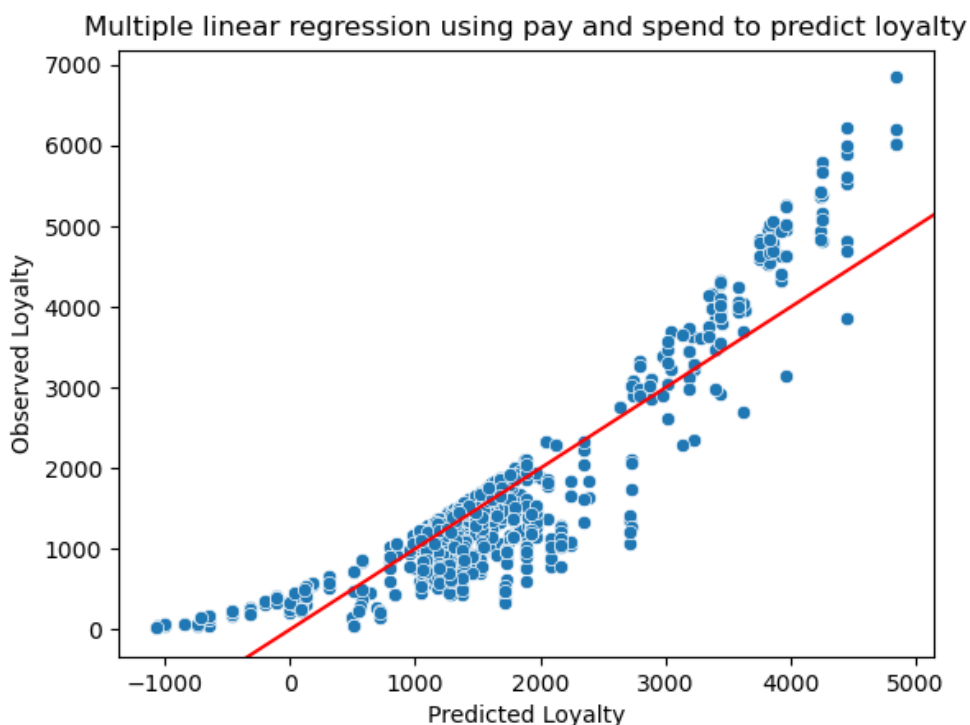
Histograms and boxplots of loyalty points indicated that most customers earned a moderate number of points, with a smaller proportion accumulating much higher values. This suggests that an income-based tiered loyalty structure could be effective: rewarding high-value customers while motivating the less engaged customers with introductory incentives. Scatterplots showed a positive correlation between pay and loyalty points, indicating higher-income groups are more engaged in the programme.

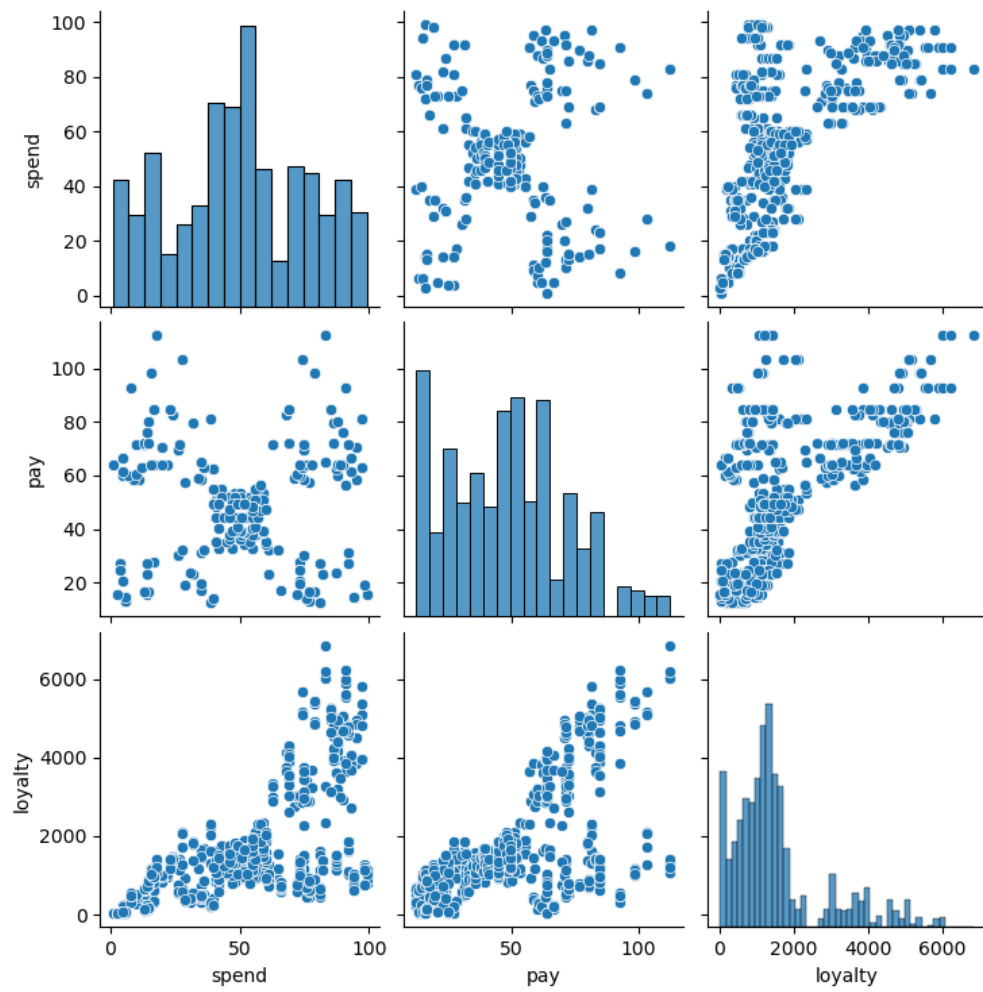


### 3.2 Regression Analysis

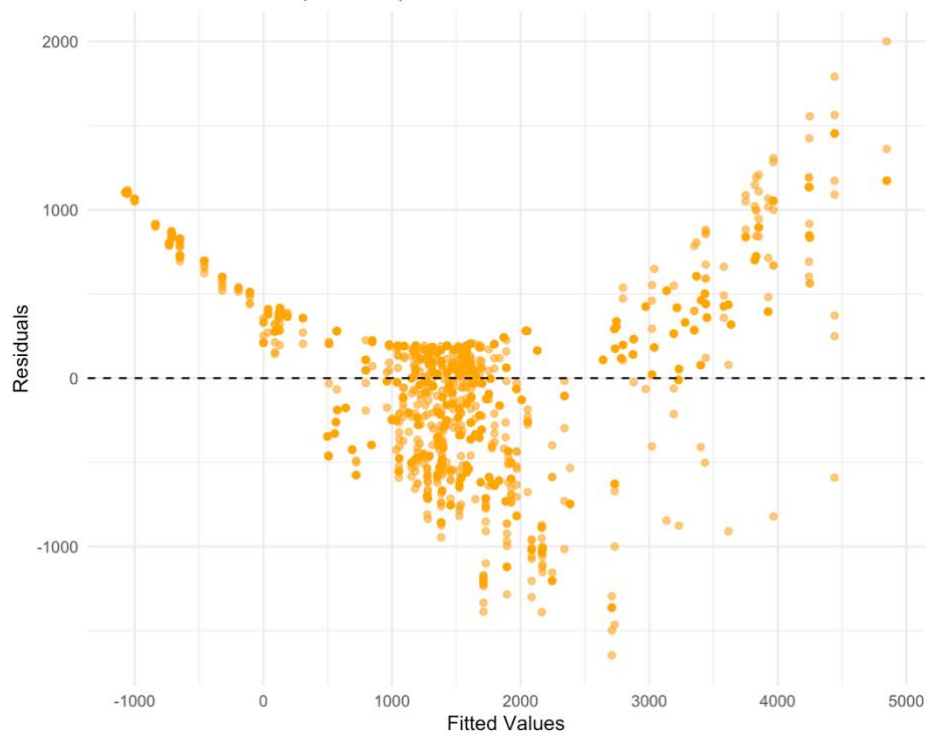
Regression plots demonstrated clear positive relationships between loyalty points and both remuneration and spending. Multiple linear regression outputs confirmed remuneration and spending score as significant predictors, with age showing a weaker influence. The adjusted  $R^2$  (0.830) indicated that these factors jointly explained 83% of the variation in loyalty points. This provides a more reliable predictive foundation than single-variable models and indicates a strong fit, reinforcing the importance of financial capacity as a predictor of customer engagement.

| OLS Regression Results |                  |         |                     |           |           |           |
|------------------------|------------------|---------|---------------------|-----------|-----------|-----------|
| =====                  |                  |         |                     |           |           |           |
| Dep. Variable:         | loyalty          |         | R-squared:          | 0.830     |           |           |
| Model:                 | OLS              |         | Adj. R-squared:     | 0.830     |           |           |
| Method:                | Least Squares    |         | F-statistic:        | 3895.     |           |           |
| Date:                  | Sat, 04 Oct 2025 |         | Prob (F-statistic): | 0.00      |           |           |
| Time:                  | 09:25:55         |         | Log-Likelihood:     | -12307.   |           |           |
| No. Observations:      | 1600             |         | AIC:                | 2.462e+04 |           |           |
| Df Residuals:          | 1597             |         | BIC:                | 2.464e+04 |           |           |
| Df Model:              | 2                |         |                     |           |           |           |
| Covariance Type:       | nonrobust        |         |                     |           |           |           |
| =====                  |                  |         |                     |           |           |           |
|                        | coef             | std err | t                   | P> t      | [0.025    | 0.975]    |
| =====                  |                  |         |                     |           |           |           |
| const                  | -1700.3237       | 39.588  | -42.950             | 0.000     | -1777.974 | -1622.674 |
| spend                  | 32.6439          | 0.510   | 63.947              | 0.000     | 31.643    | 33.645    |
| pay                    | 34.3346          | 0.574   | 59.838              | 0.000     | 33.209    | 35.460    |
| =====                  |                  |         |                     |           |           |           |
| Omnibus:               | 2.977            |         | Durbin-Watson:      | 2.034     |           |           |
| Prob(Omnibus):         | 0.226            |         | Jarque-Bera (JB):   | 2.923     |           |           |
| Skew:                  | 0.075            |         | Prob(JB):           | 0.232     |           |           |
| Kurtosis:              | 3.147            |         | Cond. No.           | 220.      |           |           |
| =====                  |                  |         |                     |           |           |           |





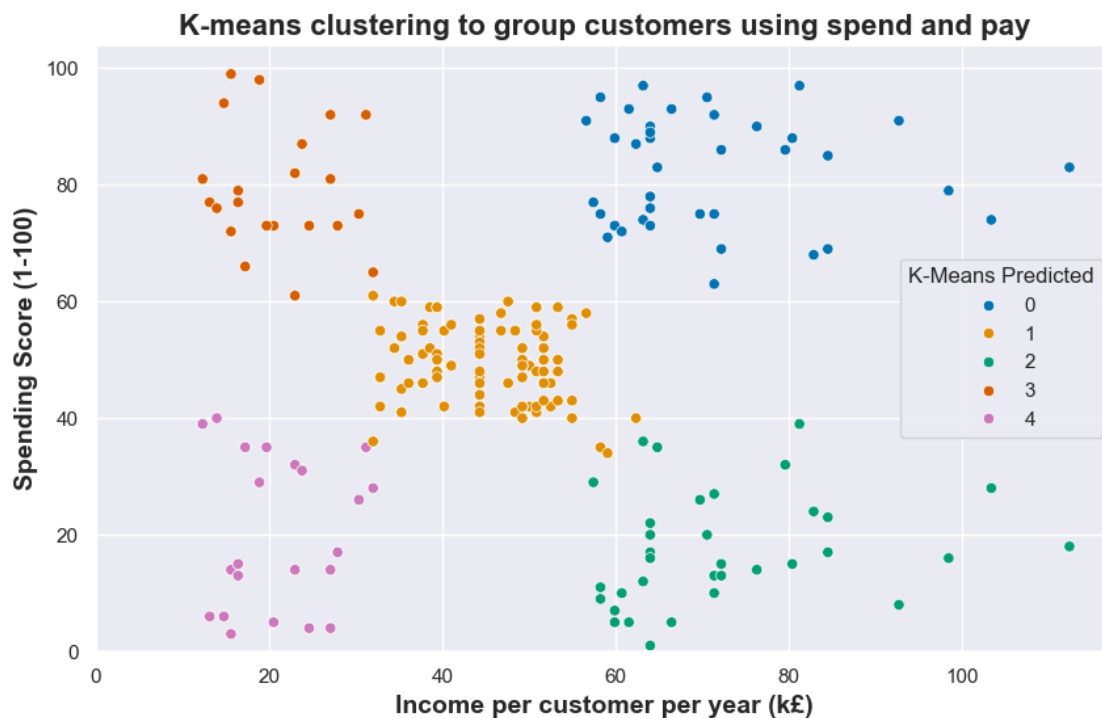
Residuals vs Fitted (Model C)



### 3.3 Customer Segmentation

K-means clustering identified 5 distinct groups (refer to the table). Scatterplots of spend versus pay clearly illustrated these segments. The largest group comprised moderate earners, while the most profitable group was the high-income cluster. This segmentation provides a roadmap for differentiated marketing strategies, with tailored campaigns suited to each group's spending capacity and preferences.

| Cluster Number | Income Level                   |
|----------------|--------------------------------|
| Cluster 0      | High Income, High Spending     |
| Cluster 1      | Medium Income, Medium Spending |
| Cluster 2      | High Income, Low Spending      |
| Cluster 3      | Low Income, High Spending      |
| Cluster 4      | Low Income, Low Spending       |



### 3.4 Customer Reviews and Sentiment

Sentiment analysis revealed that reviews and summaries frequently highlighted “fun,” “great,” “love,” and “five stars”, underscoring Turtle Games’ strength in offering social, affordable entertainment. Negative reviews frequently referenced the word “quality”, indicating areas for product improvement and quality assurance processes.

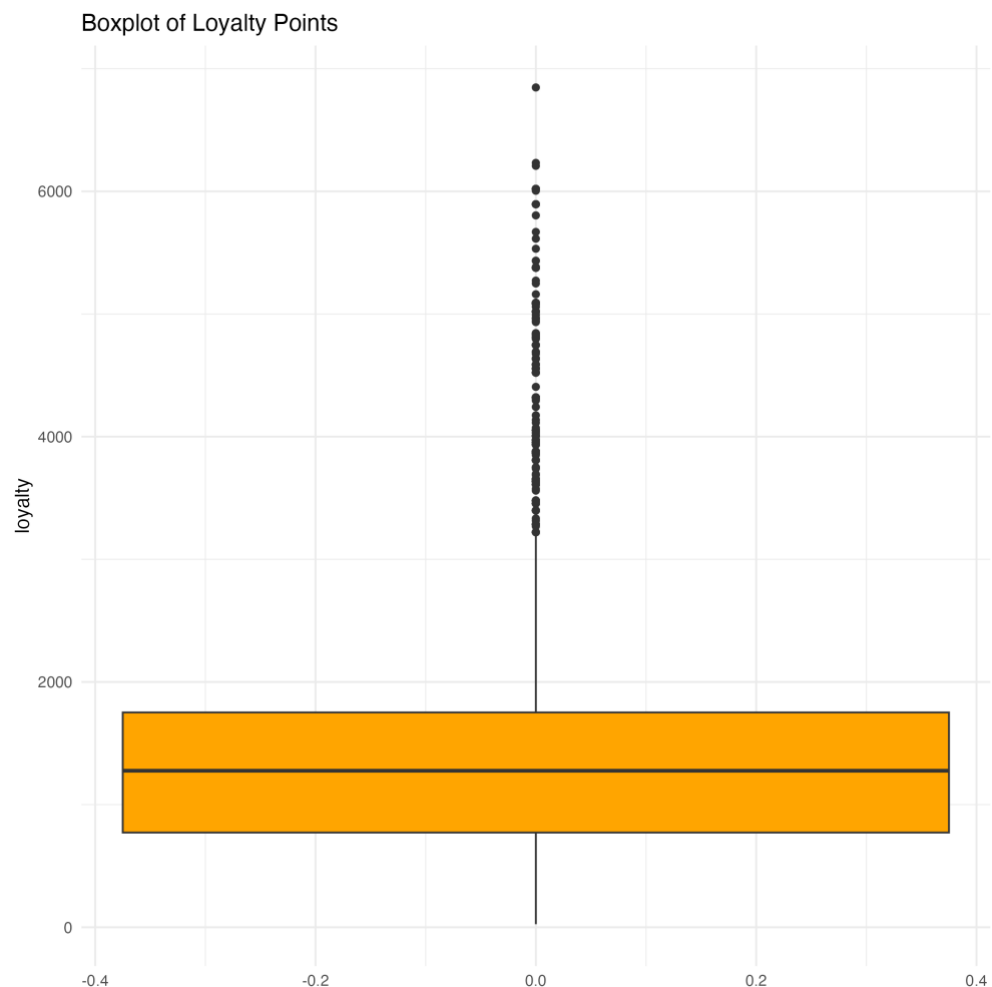
Histograms of sentiment polarity confirmed that the majority of reviews were positive, with a smaller but significant portion expressing dissatisfaction with the quality of the product. This indicates strong brand perception but highlights an urgent need to improve the quality of the product.



## 4. Patterns and Predictions

### 4.1 Financial Capacity as a Key Driver

- Customers with **higher remuneration** and **spending scores** consistently earned more loyalty points.
- Customers with low remuneration and relatively high spending scores formed a “stretch spender” group; although smaller, they show strong engagement potential if incentivised effectively. Hence, this group could be the prime target for bundle promotions.



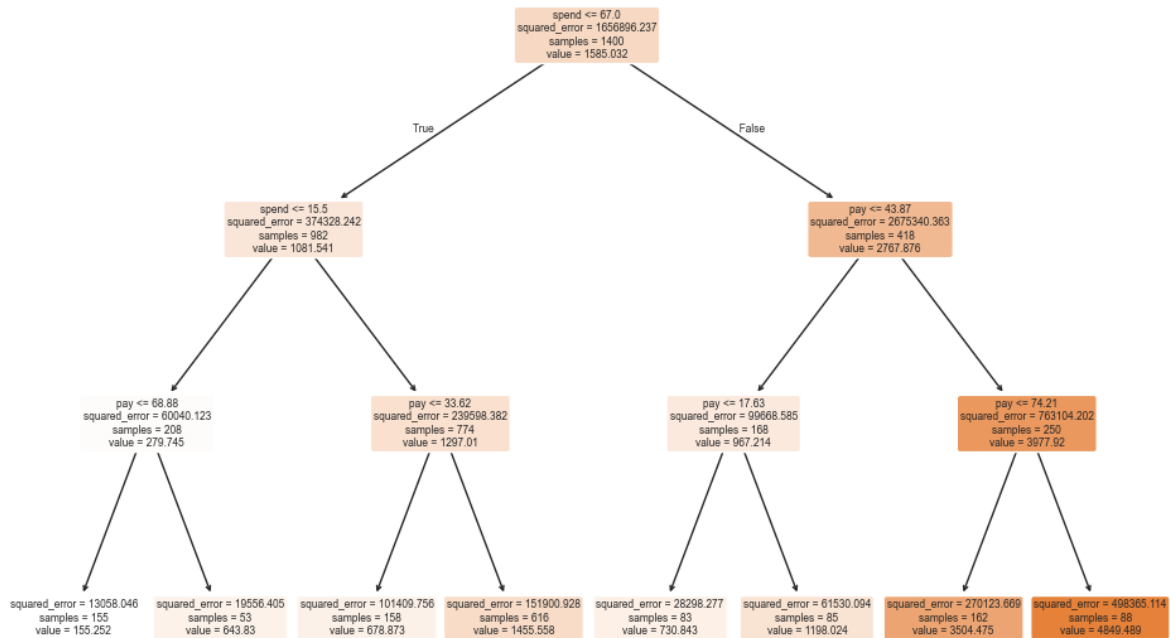
### 4.2 Multiple Linear Regression Insights

- Pay and Spend were strong, significant predictors of loyalty points.
- Age showed a weak relationship.

### 4.3 Decision Tree Regressor

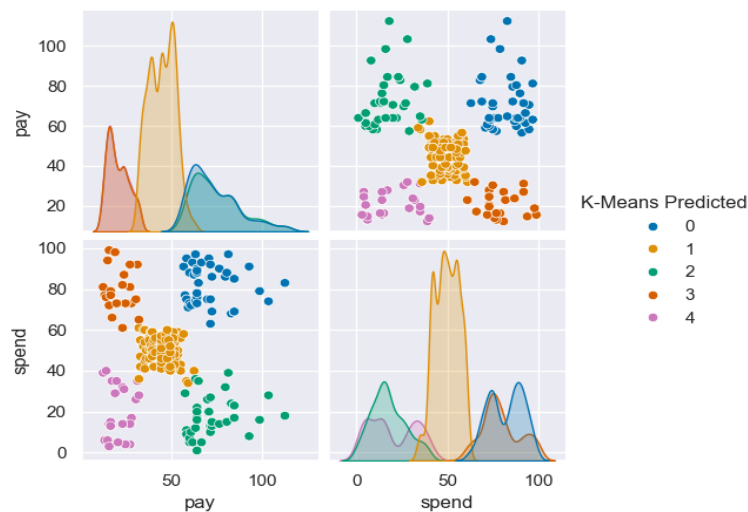
- Segmented customers based on spending and income
- Pruned tree balances simplicity and predictive accuracy, avoiding overfitting while remaining interpretable.

- Leaves show average loyalty points for customer segments.



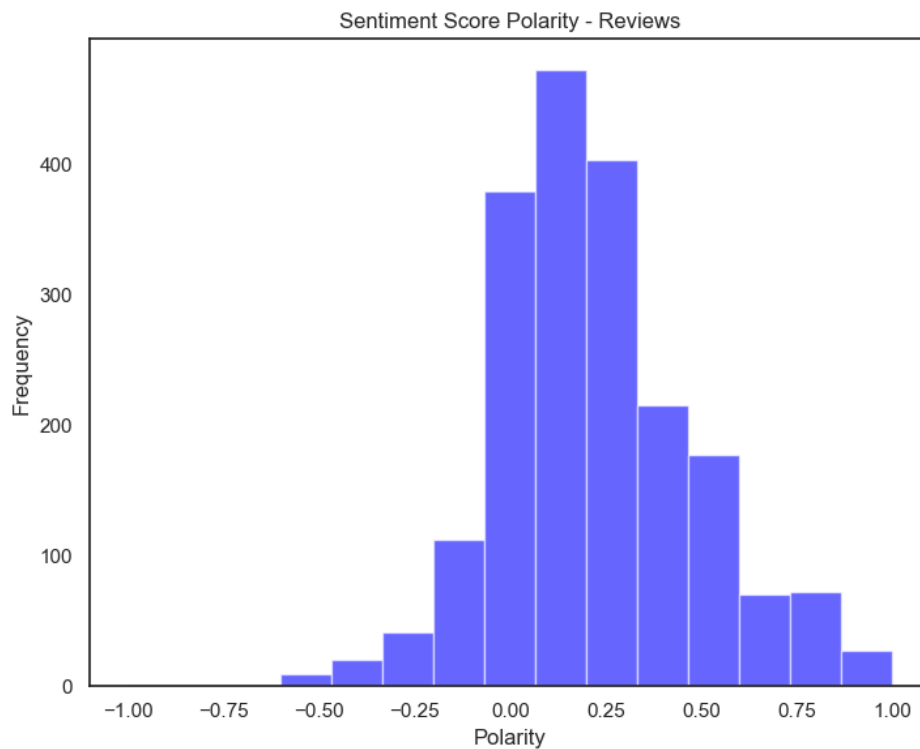
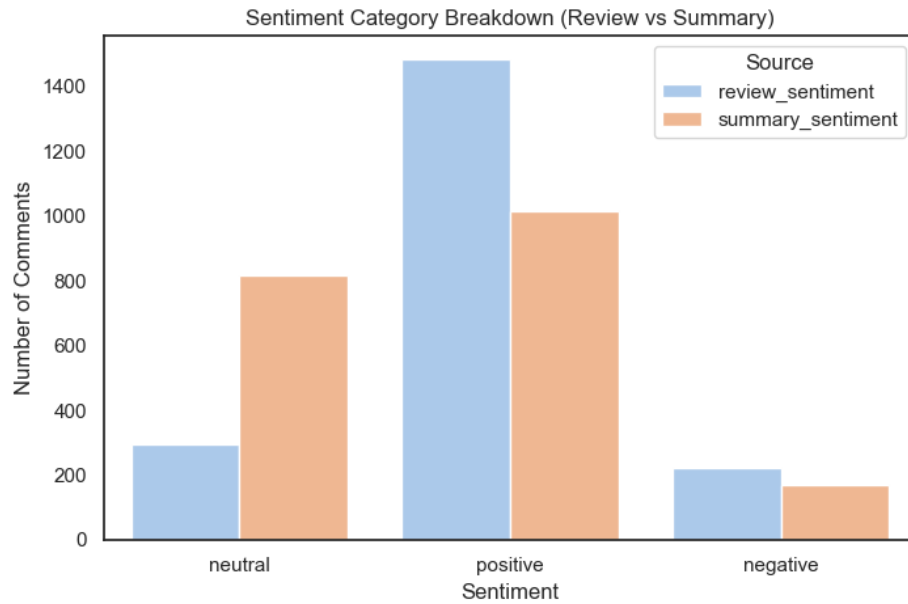
#### 4.4 Customer Segmentation (Clustering)

- K-means identified 5 clusters:
  - High-income/High-spending (small but highly profitable).
  - Moderate earners with balanced spending (largest group).
  - Low-income/Low-spending (least profitable)



#### 4.5 Sentiment Analysis and Loyalty

- Positive reviews correlated with higher loyalty engagement.
- Negative reviews highlight the bad quality of the product.



#### 4.6 Predictive Value of Models

- Financial and behavioural factors reliably forecast loyalty points.
- Regression, decision trees and clustering provide complementary insights for targeted marketing.

### 5. Recommendations for Business

- **Tiered Loyalty Programme** – Reward high-engagement customers with premium benefits while motivating fewer active ones with entry-level rewards.
- **Targeted Marketing** – Focus campaigns on high-income, high-spending groups; offer bundles for moderate earners and affordable deals for low-income segments.
- **Operational Improvements** – Focus on the improvement of product quality, stock availability and delivery to reduce customer dissatisfaction.
- **Predictive Modelling** – Use regression, clustering, and decision trees in marketing to forecast loyalty and guide incentives.
- **Analytical Improvements** – Integrate categorical features (gender, education, etc) to refine segmentation and predictions.

Implementing these strategies will boost loyalty, satisfaction and sales growth.

## Appendix

### A1. Regression Analysis Outputs

#### A1.1 Regression Coefficients Table

| OLS Regression Results |                  |                     |           |       |           |           |
|------------------------|------------------|---------------------|-----------|-------|-----------|-----------|
| Dep. Variable:         | loyalty          | R-squared:          | 0.830     |       |           |           |
| Model:                 | OLS              | Adj. R-squared:     | 0.830     |       |           |           |
| Method:                | Least Squares    | F-statistic:        | 3895.     |       |           |           |
| Date:                  | Sat, 04 Oct 2025 | Prob (F-statistic): | 0.00      |       |           |           |
| Time:                  | 10:24:53         | Log-Likelihood:     | -12307.   |       |           |           |
| No. Observations:      | 1600             | AIC:                | 2.462e+04 |       |           |           |
| Df Residuals:          | 1597             | BIC:                | 2.464e+04 |       |           |           |
| Df Model:              | 2                |                     |           |       |           |           |
| Covariance Type:       | nonrobust        |                     |           |       |           |           |
|                        | coef             | std err             | t         | P> t  | [0.025    | 0.975]    |
| const                  | -1700.3237       | 39.588              | -42.950   | 0.000 | -1777.974 | -1622.674 |
| spend                  | 32.6439          | 0.510               | 63.947    | 0.000 | 31.643    | 33.645    |
| pay                    | 34.3346          | 0.574               | 59.838    | 0.000 | 33.209    | 35.460    |
| Omnibus:               | 2.977            | Durbin-Watson:      | 2.034     |       |           |           |
| Prob(Omnibus):         | 0.226            | Jarque-Bera (JB):   | 2.923     |       |           |           |
| Skew:                  | 0.075            | Prob(JB):           | 0.232     |       |           |           |
| Kurtosis:              | 3.147            | Cond. No.           | 220.      |       |           |           |

#### A1.2 Descriptive Statistics Table

- Spend vs Loyalty Points:

| OLS Regression Results |                  |                   |                     |           |          |        |
|------------------------|------------------|-------------------|---------------------|-----------|----------|--------|
| Dep. Variable:         | loyalty          |                   | R-squared:          | 0.452     |          |        |
| Model:                 | OLS              |                   | Adj. R-squared:     | 0.452     |          |        |
| Method:                | Least Squares    |                   | F-statistic:        | 1648.     |          |        |
| Date:                  | Sat, 04 Oct 2025 |                   | Prob (F-statistic): | 2.92e-263 |          |        |
| Time:                  | 14:16:19         |                   | Log-Likelihood:     | -16550.   |          |        |
| No. Observations:      | 2000             |                   | AIC:                | 3.310e+04 |          |        |
| Df Residuals:          | 1998             |                   | BIC:                | 3.312e+04 |          |        |
| Df Model:              | 1                |                   |                     |           |          |        |
| Covariance Type:       | nonrobust        |                   |                     |           |          |        |
|                        | coef             | std err           | t                   | P> t      | [0.025   | 0.975] |
| Intercept              | -75.0527         | 45.931            | -1.634              | 0.102     | -165.129 | 15.024 |
| spend                  | 33.0617          | 0.814             | 40.595              | 0.000     | 31.464   | 34.659 |
| Omnibus:               | 126.554          | Durbin-Watson:    | 1.191               |           |          |        |
| Prob(Omnibus):         | 0.000            | Jarque-Bera (JB): | 260.528             |           |          |        |
| Skew:                  | 0.422            | Prob(JB):         | 2.67e-57            |           |          |        |
| Kurtosis:              | 4.554            | Cond. No.         | 122.                |           |          |        |

- Pay vs Loyalty Points:

| OLS Regression Results |                  |                   |                     |           |          |        |
|------------------------|------------------|-------------------|---------------------|-----------|----------|--------|
| Dep. Variable:         | loyalty          |                   | R-squared:          | 0.380     |          |        |
| Model:                 | OLS              |                   | Adj. R-squared:     | 0.379     |          |        |
| Method:                | Least Squares    |                   | F-statistic:        | 1222.     |          |        |
| Date:                  | Sat, 04 Oct 2025 |                   | Prob (F-statistic): | 2.43e-209 |          |        |
| Time:                  | 14:16:48         |                   | Log-Likelihood:     | -16674.   |          |        |
| No. Observations:      | 2000             |                   | AIC:                | 3.335e+04 |          |        |
| Df Residuals:          | 1998             |                   | BIC:                | 3.336e+04 |          |        |
| Df Model:              | 1                |                   |                     |           |          |        |
| Covariance Type:       | nonrobust        |                   |                     |           |          |        |
|                        | coef             | std err           | t                   | P> t      | [0.025   | 0.975] |
| Intercept              | -65.6865         | 52.171            | -1.259              | 0.208     | -168.001 | 36.628 |
| pay                    | 34.1878          | 0.978             | 34.960              | 0.000     | 32.270   | 36.106 |
| Omnibus:               | 21.285           | Durbin-Watson:    | 3.622               |           |          |        |
| Prob(Omnibus):         | 0.000            | Jarque-Bera (JB): | 31.715              |           |          |        |
| Skew:                  | 0.089            | Prob(JB):         | 1.30e-07            |           |          |        |
| Kurtosis:              | 3.590            | Cond. No.         | 123.                |           |          |        |

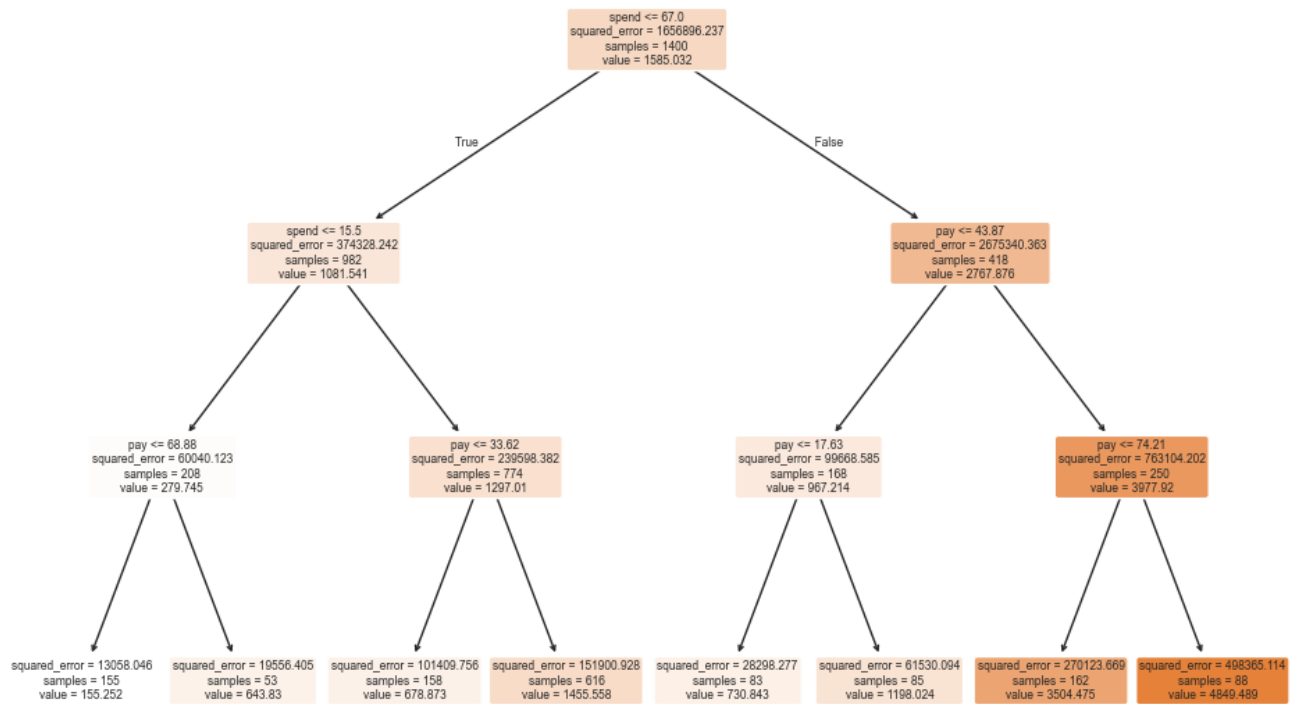
- Age vs Loyalty Points:

| OLS Regression Results |                  |                   |                     |           |          |          |
|------------------------|------------------|-------------------|---------------------|-----------|----------|----------|
| Dep. Variable:         | loyalty          |                   | R-squared:          | 0.002     |          |          |
| Model:                 | OLS              |                   | Adj. R-squared:     | 0.001     |          |          |
| Method:                | Least Squares    |                   | F-statistic:        | 3.606     |          |          |
| Date:                  | Sat, 04 Oct 2025 |                   | Prob (F-statistic): | 0.0577    |          |          |
| Time:                  | 10:24:47         |                   | Log-Likelihood:     | -17150.   |          |          |
| No. Observations:      | 2000             |                   | AIC:                | 3.430e+04 |          |          |
| Df Residuals:          | 1998             |                   | BIC:                | 3.431e+04 |          |          |
| Df Model:              | 1                |                   |                     |           |          |          |
| Covariance Type:       | nonrobust        |                   |                     |           |          |          |
|                        | coef             | std err           | t                   | P> t      | [0.025   | 0.975]   |
| Intercept              | 1736.5177        | 88.249            | 19.678              | 0.000     | 1563.449 | 1909.587 |
| age                    | -4.0128          | 2.113             | -1.899              | 0.058     | -8.157   | 0.131    |
| Omnibus:               | 481.477          | Durbin-Watson:    | 2.277               |           |          |          |
| Prob(Omnibus):         | 0.000            | Jarque-Bera (JB): | 937.734             |           |          |          |
| Skew:                  | 1.449            | Prob(JB):         | 2.36e-204           |           |          |          |
| Kurtosis:              | 4.688            | Cond. No.         | 129.                |           |          |          |

## A2. Decision Trees and Random Forest Models

### A2.1 Decision Tree Visualisation

- Pruned Decision Tree predicting loyalty points based on pay and spend. Leaves indicate average loyalty points per segment.



## A2.2 Decision Tree Metrics

- Mean squared error (MSE), R2 score and depth of tree before/after pruning.

```
# Predict the response for the data test.
y_predict = regressor.predict(X_test)

# predict on test data and evaluate performance before pruning
print("Before pruning:")
print("MSE:", mean_squared_error(y_test, y_predict))
print("R2 Score:", r2_score(y_test, y_predict))

Before pruning:
MSE: 26097.98366433656
R2 Score: 0.9838881140581562

# Prune the model.
# Apply basic pruning (limit tree depth and min_samples_leaf)
pruned_regressor = DecisionTreeRegressor(
    random_state=42,
    max_depth=3,          # maximum depth of tree
    min_samples_leaf=5    # minimum samples per leaf
)
pruned_regressor.fit(X_train, y_train)

DecisionTreeRegressor
DecisionTreeRegressor(max_depth=3, min_samples_leaf=5, random_state=42)

# Evaluate after pruning
y_predict_pruned = pruned_regressor.predict(X_test)

print("\nAfter pruning:")
print("MSE:", mean_squared_error(y_test, y_predict_pruned))
print("R2 Score:", r2_score(y_test, y_predict_pruned))

After pruning:
MSE: 137962.67325147352
R2 Score: 0.9148271803581075
```

## A2.3 Random Forest Code

### 1. Load and Prepare Data

```
[210]: # Load and prepare data
turtle = pd.read_csv('turtle_reviews_clean.csv')
turtle = turtle.drop('Unnamed: 0', axis=1)

# View DataFrame
turtle
```

| [210]: | gender | age | pay   | spend | loyalty | education | product | review                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | summary                                                                                          |
|--------|--------|-----|-------|-------|---------|-----------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 0      | Male   | 18  | 12.30 | 39    | 210     | graduate  | 453     | When it comes to a DM's screen, the space on the screen itself is at an absolute premium. The fact that 50% of this space is wasted on art (and not terribly informative or needed art as well) makes it completely useless. The only reason that I gave it 2 stars and not 1 was that, technically speaking, it can at least still stand up to block your notes and dice rolls. Other than that, it drops the ball completely.                                                                                                                                                                                                                                                                                                                                                                    | The fact that 50% of this space is wasted on art (and not terribly informative or needed art ... |
| 1      | Male   | 23  | 12.30 | 81    | 524     | graduate  | 466     | An Open Letter to GaleForce9:<br>Your unpainted miniatures are very not bad. Your spell cards are great. Your board games are "meh". Your DM screens, however, are freaking terrible. I'm still waiting for a single screen that isn't polluted with pointless artwork where useful, reference-able tables should be. Once again, you've created a single use screen that is only useful when running the "Storm King's Thunder" adventure. Even despite the fact that it's geared to that adventure path, it's usefulness negligible, at best. I massive swath of the inner panel is wasted on artwork and a bloated overland map, which could have been easily reduced to a single panel in size. And the few table you have are nigh-useless themselves.<br>short, stop making crap DM screens. | Another worthless Dungeon Master's screen from GaleForce9                                        |

```
[211]: # Define features and target variable
X = turtle.drop(['loyalty', 'review', 'summary'], axis=1)
y = turtle['loyalty']

[212]: # Encode categorical variables
for col in ['gender', 'education']:
    X[col] = LabelEncoder().fit_transform(X[col])

[213]: # Split data (70% training, 30% testing)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

[214]: # Train model
rf_model = RandomForestRegressor(n_estimators=100, max_depth=10, random_state=42, n_jobs=-1)
rf_model.fit(X_train, y_train)
```

RandomForestRegressor

RandomForestRegressor(max\_depth=10, n\_jobs=-1, random\_state=42)

```
[215]: # Predictions
y_train_pred = rf_model.predict(X_train)
y_test_pred = rf_model.predict(X_test)

[216]: # Evaluation metrics
print("Training Set:")
print(f" R² Score: {r2_score(y_train, y_train_pred):.4f}")
print(f" RMSE: {np.sqrt(mean_squared_error(y_train, y_train_pred)):.2f}")
print(f" MAE: {mean_absolute_error(y_train, y_train_pred):.2f}")

print("\nTest Set:")
print(f" R² Score: {r2_score(y_test, y_test_pred):.4f}")
print(f" RMSE: {np.sqrt(mean_squared_error(y_test, y_test_pred)):.2f}")
print(f" MAE: {mean_absolute_error(y_test, y_test_pred):.2f}")

Training Set:
R² Score: 0.9982
RMSE: 55.20
MAE: 27.82

Test Set:
R² Score: 0.9947
RMSE: 92.39
MAE: 46.07

[217]: # Feature importance
importance_df = pd.DataFrame({
    'Feature': X.columns,
    'Importance': rf_model.feature_importances_
}).sort_values('Importance', ascending=False)
print("\nFeature Importance:")
print(importance_df)
```

```
[217]: # Feature importance
importance_df = pd.DataFrame({
    'Feature': X.columns,
    'Importance': rf_model.feature_importances_
}).sort_values('Importance', ascending=False)
print("\nFeature Importance:")
print(importance_df)
```

|   | Feature   | Importance |
|---|-----------|------------|
| 3 | spend     | 0.499975   |
| 2 | pay       | 0.482842   |
| 1 | age       | 0.014528   |
| 5 | product   | 0.001370   |
| 4 | education | 0.000840   |
| 0 | gender    | 0.000445   |

```
[218]: # Prediction accuracy
errors = np.abs(y_test - y_test_pred)
print(f"\nPredictions within ±100 points: {(errors <= 100).sum() / len(y_test) * 100:.2f}%")
print(f"Predictions within ±200 points: {(errors <= 200).sum() / len(y_test) * 100:.2f}%")

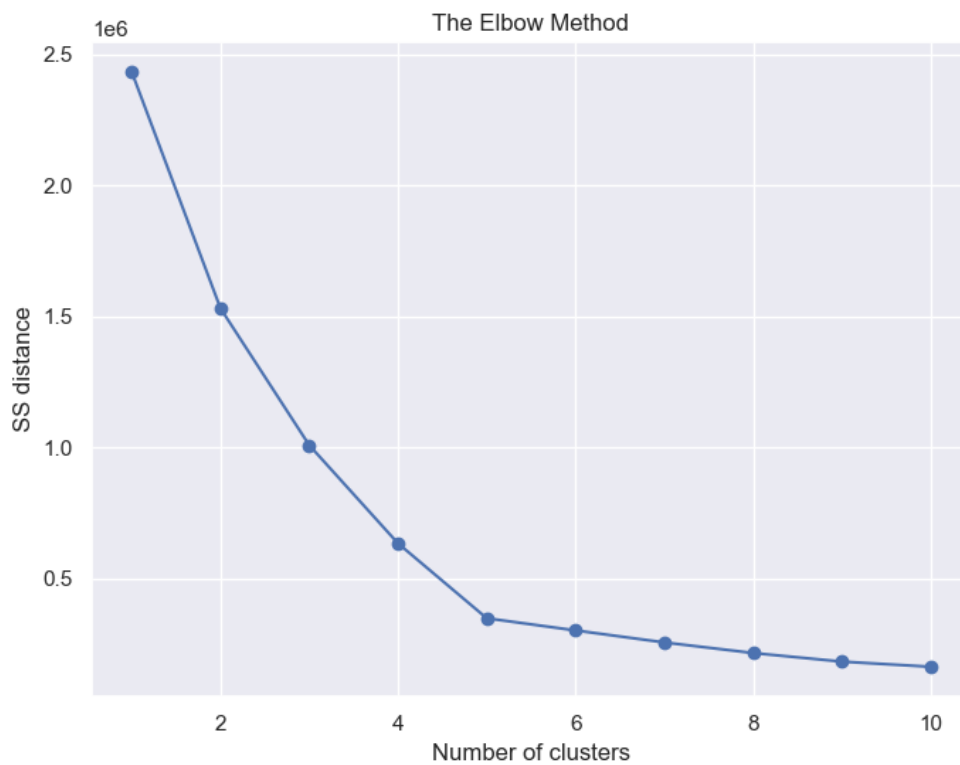
Predictions within ±100 points: 84.67%
Predictions within ±200 points: 95.50%
```

## A2.4 Random Forest Insights

| Feature                | Importance |
|------------------------|------------|
| Spending Score (Spend) | 0.50       |
| Remuneration (Pay)     | 0.48       |
| Age                    | 0.01       |

## A3. Clustering Outputs

### A3.1 Elbow Method Graph



- The Elbow Method plot suggests that 5 is the most appropriate k value to use - this is where there is a slight but obvious kink in the graph.

### A3.2 Silhouette Score Plot



- The silhouette method strongly suggests that 5 is the correct number of clusters, as there is an obvious peak in the line plot at k=5. This is supported by the Elbow Method findings.

#### A4. Sentiment Analysis Outputs

### A4.1 Word Clouds

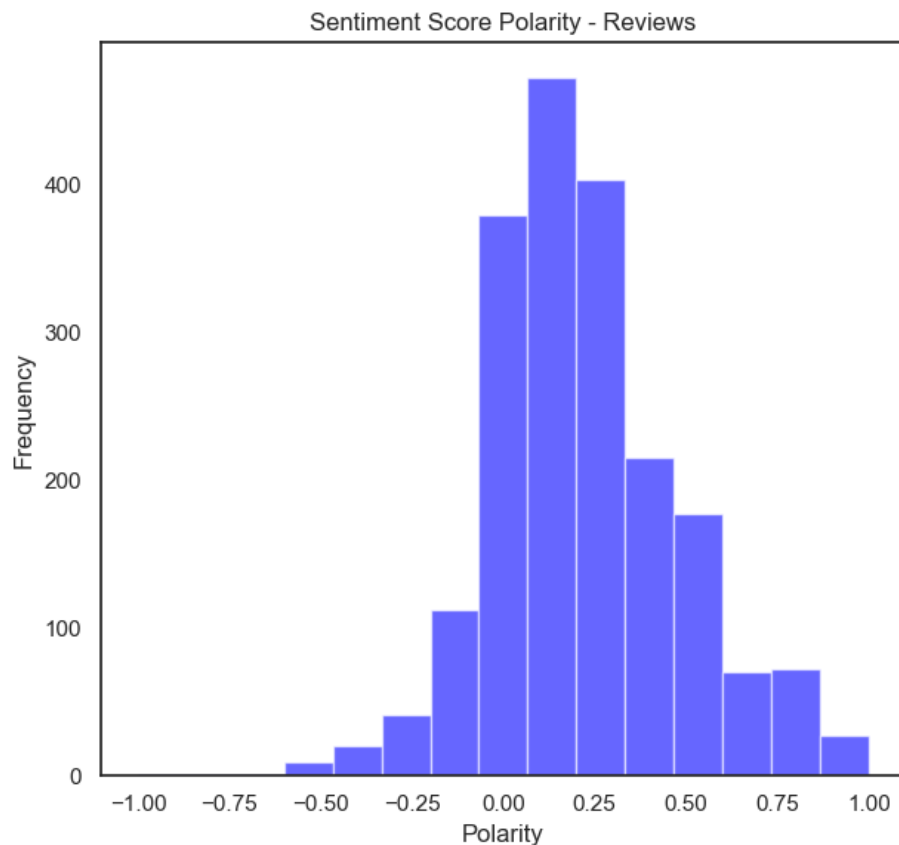
### Reviews Word Cloud:



### Summary Word Cloud:

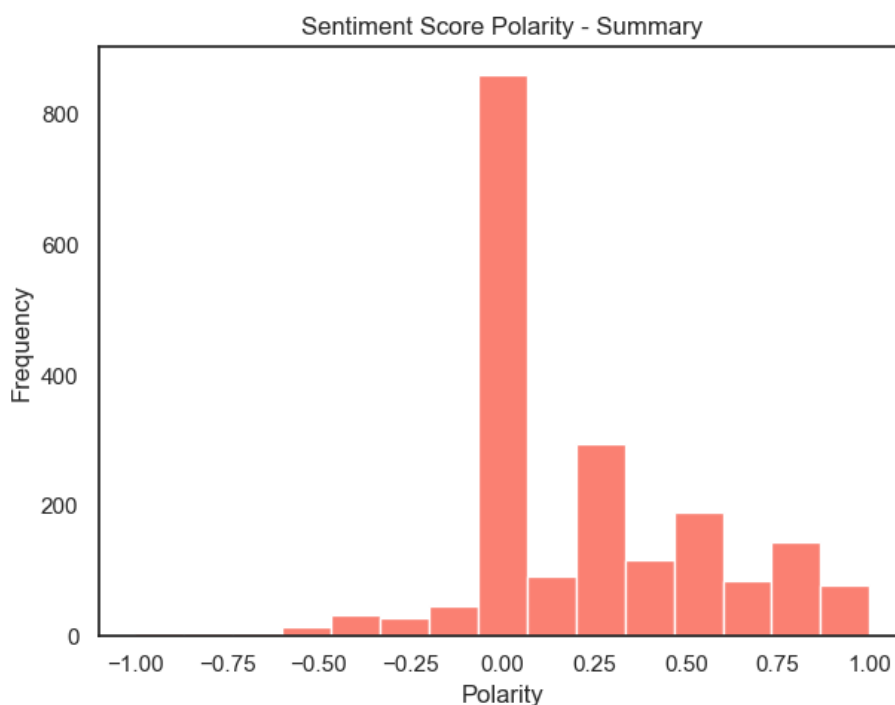


## A4.2 Sentiment Polarity Histograms



- The distribution appears to be slightly right-skewed, centred around mild positivity near 0.15-0.25, indicating that most reviews express modestly positive sentiment.

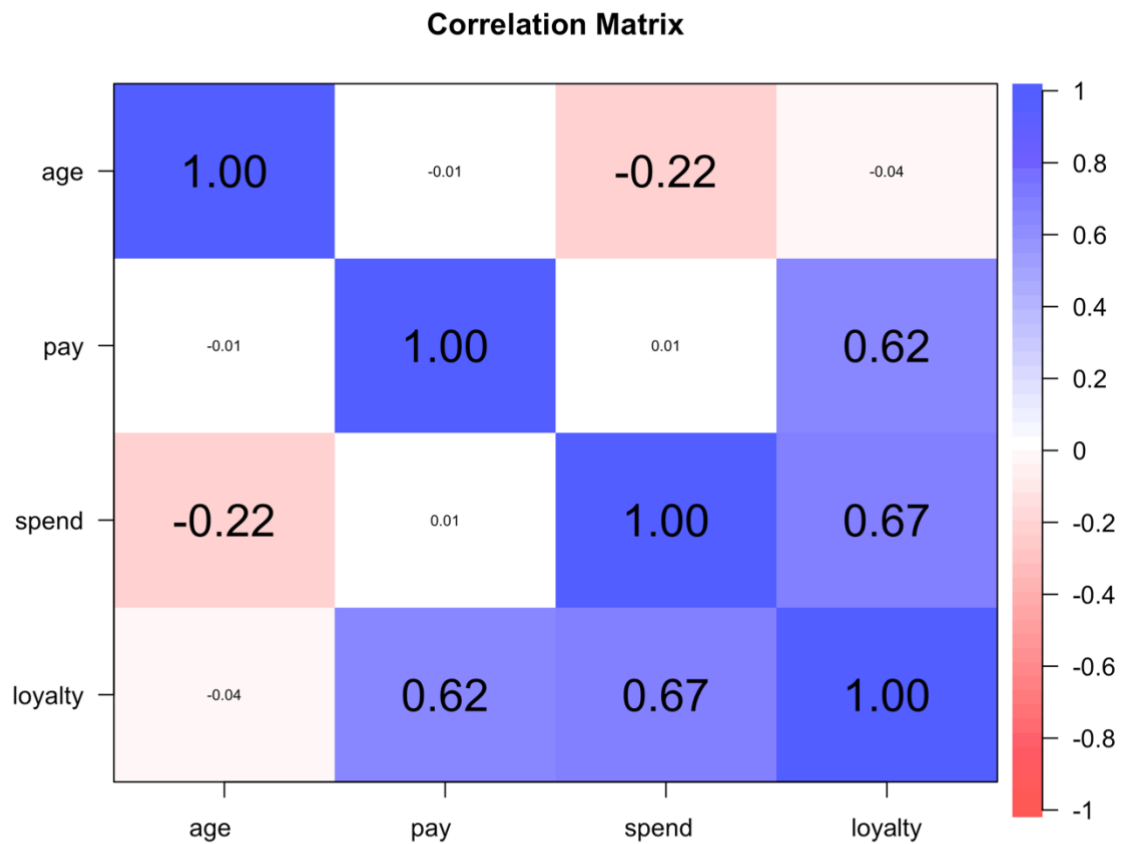
- A long, thinner tail extends into high positivity up to about, suggesting a smaller subset of very enthusiastic reviews; the left tail into negatives is shorter and sparser, implying fewer strongly negative experiences



- Most summaries cluster tightly around neutral-to-slightly-positive polarity with a strong positive skew, indicating short texts that are cautious in tone overall but with a meaningful tail of enthusiastic entries.
- The very tall bin near 0 suggests a high prevalence of neutral or weakly positive wording, a common outcome when short summaries compress nuance and avoid extreme language, consistent with polarity scales where 0 is neutral and +1 is very positive.

## A5. Additional Supporting Visuals

### A5.1 Correlation Matrix of Variables (Pay, Spend, Loyalty, Age)



#### Correlations:

- Strongest positive correlation: spend and loyalty (0.67), indicating that higher spending customers are significantly more loyal
- Moderate positive correlation: pay and loyalty (0.62), suggesting customers with higher payment values exhibit greater loyalty
- Negative correlation: age and spend (-0.22), showing that younger customers tend to spend more than older customers
- Pay-spend-loyalty triangle forms a reinforcing cycle, while age operates independently with minimal impact on loyalty

**Spending is the strongest loyalty driver (0.67).**

## A5.2 Comparative Table of all Predictive Models

| Model               | Key Variables      | Strengths                     | Limitations                        |
|---------------------|--------------------|-------------------------------|------------------------------------|
| Linear Regression   | Pay, Spend         | Easy to interpret, high $R^2$ | Sensitive to outliers              |
| Multiple Regression | Pay, Spend, Age    | Stronger predictive validity  | Age less impactful                 |
| Decision Tree       | Pay, Spend         | Easy segmentation             | Risk of overfitting                |
| Random Forest       | Pay, Spend, Age    | Robust, feature importance    | Less interpretable than regression |
| K-Means Clustering  | Pay, Spend         | Clear customer segments       | Clusters uneven in size            |
| Sentiment Analysis  | Reviews, Summaries | Captures customer perceptions | Limited to text, subjective bias   |